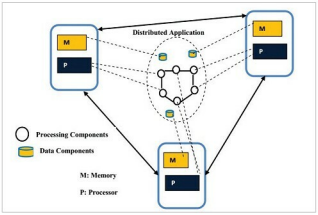

Figure 1: Distributed computing environment

needs, and then compile them using the gcc compiler as shown in Figure 5.

```
/*  This is  dist_app_client.c generated by rpcgen and
modified according to the need */

#include “dist_app.h”
void minmax_1(char *host1,char *host2,int *p,int length) //
Client Program has to communicate
{                    // with two Computers as per Figure 2.
     CLIENT *clnt1,*clnt2;
     int  *result_1,i;
     struct ipair op;
     int  *result_2;
     for(i=0;i<length;i++)            //making data ready
           op.data[i]=p[i];
           op.count=length;

#ifndef DEBUG
     clnt1 = clnt_create (host1, MINMAX, VER1, “udp”);  //
for making UDP connection
     if (clnt1 == NULL) {
       clnt_pcreateerror (host1);
       exit (1);
     }
     clnt2 = clnt_create (host2, MINMAX, VER1, “udp”); // for
making UDP Connection
     if (clnt2 == NULL) {
       clnt_pcreateerror (host2);
       exit (1);
     }
#endif  /* DEBUG */

     result_1 = max_1(&op, clnt1);        // sending data to
service Max as per Figure 2
     if (result_1 == (int *) NULL) {
       clnt_perror (clnt1, “call failed”);
     }
          printf(“\n The result from the Service Max is
```

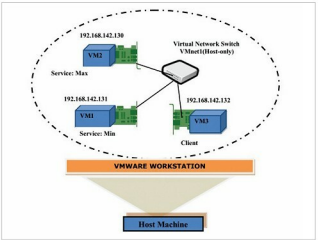

Figure 2: Test bed

```
root@client-virtual-machine: /home/client/RPC
root@client-virtual-machine:/home/client/RPC# rpcgen -C dist_app.x
root@client-virtual-machine:/home/client/RPC# ls
dist_app_clnt.c  dist_app.h  dist_app_svc.c  dist_app.x  dist_app_xdr.c
```

Figure 3: Compiling the IDL file with *rpcgen*

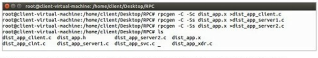

Figure 4: Creating sample client and sample server programs with *rpcgen*

Figure 5: Generating a distributed application with gcc

```
:%d\n”, *result_1);
     result_2 = min_1(&op, clnt2);       // sending data to
service Min as per Figure 2
     if (result_2 == (int *) NULL) {
       clnt_perror (clnt2, “call failed”);
     }
          printf(“\n The result from the Service Min is
:%d\n”, *result_2);
#ifndef DEBUG
     clnt_destroy (clnt1);
          clnt_destroy (clnt2);
#endif  /* DEBUG */
}
int main (int argc, char *argv[])
{
     char *host1,*host2;

     int i,x[10];
          printf(“\n give any 10 elements”);  // For
collecting the list of elements from USER
                    for(i=0;i<10;i++)
                    scanf(“%d”,&x[i]);
```